

Algebra Quiz

Question 1

Which of the following pairs consists of like terms?

- ☐ $5x^2y$ and $-2xy^2$
- ☐ $3ab$ and $3bc$
- ☒ $-7m^2n$ and $4nm^2$
- ☐ $8x$ and 8

Hint: Focus on whether the variables and their respective powers are identical, ignoring the numerical coefficients.

Question 2

When combining like terms in the expression $7a^3 - 4a^3$, what is the correct result?

- ☐ 3
- ☒ $3a^3$
- ☐ $3a^0$
- ☐ $11a^3$

Hint: Recall the rule for combining coefficients while keeping the variable part constant.

Question 3

According to the rules for removing parentheses, what is the correct expansion of $-(2x - 5y + 3)$?

- ☐ $-2x - 5y + 3$
- ☒ $-2x + 5y - 3$
- ☐ $2x - 5y + 3$
- ☐ $-2x - 5y - 3$

Hint: If a minus sign is in front of the bracket, think about what happens to every single sign inside.

Question 4

Simplify the expression: $3(x + 2) - 2(x - 1)$

- ☒ $x + 8$
- ☐ $x + 4$
- ☐ $x + 5$
- ☐ $5x + 4$

Hint: Apply the distributive property to both parts first, paying close attention to the negative sign before the second bracket.

Question 5

Which statement about like terms is true?

- ☐ Terms must have the same coefficients to be like terms.
- ☒ Constant terms (like 5 and -12) are considered like terms.
- ☐ x^2y and xy^2 are like terms because they use the same letters.
- ☐ The order of variables in a term determines if they are like terms.

Hint: Review the specific 'Points to Note' in the section regarding constants and variable properties.

Question 6

What is the result of simplifying $2x^2 + 3x^2$?

- ☐ $5x^4$

- ☐ $6x^2$
- ☒ $5x^2$
- ☐ $5x$

Hint: Think about whether the ‘identity’ of the items (x^2) changes when you count how many of them you have.

Question 7

Identify the correct first step to simplify the expression $5a - [2a - (b - a)]$.

- ☐ $5a - [2a - b - a]$
- ☒ $5a - [2a - b + a]$
- ☐ $5a - 2a - b + a$
- ☐ $3a - (b - a)$

Hint: When dealing with nested grouping symbols, it is usually safest to work from the ‘inside out’.

Question 8

Simplify the expression $4x + 3y - 2x + y$.

- ☒ $2x + 4y$
- ☐ $6x + 4y$
- ☐ $6xy$
- ☐ $2x + 3y$

Hint: Group terms with the same variables together and perform the operations indicated by their signs.

Question 9

What happens when you simplify $3x^2 - 5x + 2 - (x^2 - 5x - 1)$?

- ☒ $2x^2 + 3$
- ☐ $2x^2 - 10x + 1$
- ☐ $4x^2 - 10x + 1$
- ☐ $2x^2 + 1$

Hint: Pay close attention to the middle term; if you have five of something and you add five of that same thing back, what is left?

Question 10

A student simplified an expression and got $5x + 2y = 7xy$. Which common error did they commit?

- ☐ Sign error in removing parentheses.
- ☒ Combining non-like terms.
- ☐ Changing the exponent incorrectly.
- ☐ Calculation error with coefficients.

Hint: Think about whether you can add ‘5 apples’ and ‘2 oranges’ to get ‘7 apple-oranges’.